

Industrial Oriented Mini Project (CM653PC)

On

"Smart Water Usage Prediction and Wastage Alert System"

Submitted

In the partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

In

Computer Science Engineering -- Artificial Intelligence And Machine Learning

by

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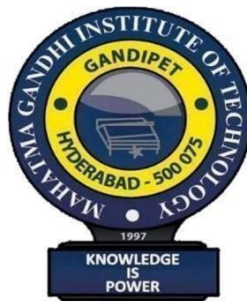
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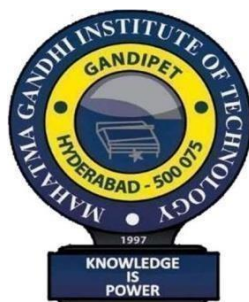
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CERTIFICATE



This is to certify that the Industrial Oriented Mini Project (CM653PC) entitled "**Smart Water Usage Prediction and Wastage Alert System**", is being submitted by **B.Sharath** bearing Roll No: **23261A6606** and **T.Sathwik** bearing Roll No: **23261A6654** in partial fulfillment for the award of degree of Bachelor of Technology in Computer Science and Engineering - Artificial Intelligence And Machine Learning to Mahatma Gandhi Institute of Technology is a record of bonafide work carried out by them under our guidance and supervision.

The results embodied in this project have not been submitted to any other University or Institute for the award of any degree or diploma.

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This is to certify that the work reported in this project titled “**Smart Water Usage Prediction and Wastage Alert System**” is a record of work done by us in the Department of Emerging Technologies, Mahatma Gandhi Institute of Technology(A), Hyderabad. No part of the work is copied from books/journals/internet and wherever the portion is taken, the same has been duly referred to in the text. The report is based on the work done entirely by us and not copied from any other source.

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ABSTRACT

Water scarcity is a major challenge in the present scenario due to rapid population growth and inefficient water usage. Proper monitoring and prediction of water consumption are essential to reduce wastage and promote sustainable living.

This project presents a Smart Water Usage Prediction and Wastage Alert System using Machine Learning. The system accepts input parameters such as the number of people in a household, temperature (in °C), and type of day (weekday/weekend) to estimate daily water consumption. A Linear Regression model is trained on a custom dataset of 50 records to learn the relationship between these input variables and water usage.

The predicted results are displayed through a web-based application developed using Python Flask, HTML, and CSS. Data visualization techniques including bar charts and doughnut charts (using Chart.js) are employed to present usage patterns clearly. The system automatically categorizes usage status into three levels – Efficient, Warning, or Critical – and generates alert messages when usage exceeds a predefined threshold, helping users identify and reduce wastage.

This system provides a simple, cost-effective, and user-friendly solution for household-level water management. It promotes responsible water usage and supports sustainable development.

Keywords: Water Usage Prediction, Water Conservation, Machine Learning, Linear Regression, Flask, Smart Monitoring, Sustainable Development, Data Analysis, Household Water Management.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
ML	Machine Learning
LR	Linear Regression
IoT	Internet of Things
CSV	Comma-Separated Values
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
API	Application Programming Interface
SVM	Support Vector Machine
CDN	Content Delivery Network
MGIT	Mahatma Gandhi Institute of Technology
CSE	Computer Science and Engineering
LSTM	Long Short-Term Memory
URL	Uniform Resource Locator

1. INTRODUCTION

1.1 Introduction

Water is one of the most essential natural resources for human survival and economic development. However, in recent years, water scarcity has become a serious global issue due to rapid population growth, urbanization, and inefficient usage of available water resources. Many regions across the world face frequent water shortages, making it increasingly necessary to adopt smarter and more efficient methods for water management and conservation.

Traditional water management systems mainly focus on supply management rather than on efficient usage control. They lack proper monitoring and predictive capabilities, which often leads to excessive consumption and avoidable wastage of water. In households, a significant amount of water is wasted daily due to unawareness of consumption patterns and the absence of real-time alert mechanisms.

With the rapid advancement of technology, Machine Learning has emerged as a powerful tool for analyzing large datasets and making accurate predictions. Among various algorithms, Linear Regression is widely used for predicting continuous values based on historical data.

By using factors such as the number of people in a household, temperature, and type of day (weekday or weekend), it is possible to estimate future water consumption patterns effectively.

This project proposes a Smart Water Usage Prediction and Wastage Alert System that utilizes historical data and a machine learning model to predict daily water usage. The system is implemented as a web-based application using Python Flask, HTML, CSS, and Chart.js for data visualization. It generates alerts when usage exceeds a predefined threshold, thereby encouraging responsible water usage at the household level.

1.2 Problem Definition

In many households, water consumption is not tracked on a daily basis, making it difficult for individuals to understand their usage patterns. Traditional water management approaches focus primarily on supply rather than efficient utilization. The lack of an intelligent, affordable, and easy-to-use household water monitoring system results in uncontrolled water consumption, unnecessary wastage, and growing environmental concerns.

The core problem addressed by this project is: How can we use Machine Learning and a lightweight web application to predict daily household water consumption and alert users when usage exceeds safe limits, without requiring expensive hardware or complex infrastructure?

1.3 Existing Systems

Several existing systems attempt to address the problem of water management. A comparative study of existing approaches is presented below:

System/Approach	Technique Used	Limitation
IoT Water Meter	Real-time sensor data	High hardware cost
Rule-based Billing	Fixed consumption thresholds	No prediction capability
Random Forest Model	Ensemble ML	Requires large datasets
SVM-based Forecast	Support Vector Machine	Complex, not user-friendly
Neural Network Forecast	Deep Learning	High compute resources needed

Table 1.1: Comparison of Existing Systems

1.4 Proposed System

The proposed Smart Water Usage Prediction and Wastage Alert System addresses the shortcomings of existing systems by integrating Machine Learning-based prediction with a web-based interface and real-time alert mechanism. The system is built using Python (Flask) for backend, HTML/CSS for frontend, Scikit-learn for the Linear Regression model, Chart.js for visualization, and a custom CSV dataset for training.

Key features include prediction of daily water usage using a trained Linear Regression model based on number of people, temperature, and day type. Usage is categorized into three status levels: Efficient (green), Warning (amber), and Critical (red). Results are displayed through an interactive Water Intelligence Dashboard with bar charts and doughnut charts.

1.5 Organization of the Report

The report is organized into eight chapters. Chapter 1 presents the introduction, problem definition, existing and proposed systems. Chapter 2 details the requirements specification.

Chapter 3 presents the literature survey. Chapter 4 describes the methodology. Chapter 5 covers the design of the system. Chapter 6 discusses the implementation. Chapter 7 presents testing, results, and discussion. Chapter 8 concludes with future scope. References, Appendix A (Source Code), and Appendix B (Steps to Execute) follow at the end.

2. REQUIREMENTS SPECIFICATION

2.1 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or above (1 GHz minimum)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 10 GB free disk space
Display	Monitor with minimum 1024x768 resolution
Network	Internet connection (for CDN libraries)

Table 2.1: Hardware

2.2 Software Requirements

Software	Details
Operating System	Windows 10/11 or Ubuntu 20.04 or above
Programming Language	Python 3.8 or above
Web Framework	Flask 2.x
ML Library	Scikit-learn 1.x
Data Handling Library	Pandas 1.x
Frontend	HTML5, CSS3, JavaScript
Visualization	Chart.js (via CDN)
Database	SQLite / CSV file
IDE / Editor	Visual Studio Code / PyCharm
Browser	Google Chrome / Mozilla Firefox

Table 2.2: Software Requirements

2.3 Functional Requirements

The system shall accept inputs: number of people (integer, 1–10), temperature in °C (integer, 15–50), and day type (weekday/weekend). It shall predict daily water consumption using a trained Linear Regression model and compare the predicted per-person usage against defined thresholds: Efficient (≤ 140 L/person), Warning (141–180 L/person), Critical (> 180 L/person). Results are displayed on an interactive dashboard with color-coded status alerts.

2.4 Non-Functional Requirements

Performance: The system shall generate predictions and display results within 2 seconds of user input submission. Usability: The interface shall be intuitive and usable without technical knowledge. Reliability: The system shall correctly predict water usage with minimal deviation from actual values. Scalability: The system shall be extensible to support multiple households and larger datasets. Portability: The system shall run on any device with a modern web browser and Python environment.

3. LITERATURE SURVEY

3.1 Review of Related Works

Water conservation and efficient management have become critical areas of research due to increasing water scarcity and rising demand. Several studies have focused on the use of advanced technologies such as Machine Learning, Internet of Things (IoT), and data analytics to monitor, predict, and optimize water usage.

Many researchers have explored the application of Machine Learning algorithms for predicting water consumption. Among these, Linear Regression is one of the most widely used techniques due to its simplicity and effectiveness in handling continuous data. Studies have shown that Linear Regression can accurately estimate water usage based on historical consumption patterns and influencing factors such as population size, weather conditions, and daily usage behavior.

In addition to Linear Regression, other algorithms such as Decision Trees, Random Forest, and Support Vector Machines have also been applied for water demand forecasting. These models provide higher accuracy in complex datasets but require more computational resources. However, for small-scale applications like household-level prediction, simpler models like Linear Regression are preferred due to ease of implementation and faster processing.

Several research works have also focused on smart water monitoring systems using IoT devices. These systems collect real-time data from sensors installed in pipelines and storage tanks to track water flow and detect leakage. While these systems provide accurate real-time monitoring, they often require expensive hardware and infrastructure, making them less suitable for individual households.

Recent studies have also emphasized the importance of web-based applications for visualizing water usage data. These platforms allow users to easily understand their consumption patterns through graphs and reports. Visualization tools help in better decisionmaking and encourage users to adopt sustainable practices.

3.2 Research Gaps

Despite the advancements in this field, many existing systems either focus only on monitoring or only on prediction. There is a lack of integrated systems that combine prediction, monitoring, and alert generation in a simple and cost-effective manner for household use. The proposed system addresses this gap by integrating Machine Learningbased prediction with a user-friendly web interface and alert mechanism.

S.No	Authors	Title	Journal	Year	Methodology	Key Findings	Dataset	Merits	Demerits
1	Proposed System	Smart Water Usage Prediction and Wastage Alert System	----	2025	Linear Regression	Predicts usage and gives alerts	Custom household dataset	Simple, low-cost, scalable	Limited to small dataset
2	Chenetal.	Forecasting Water Consumption Using ML	Elsevier Sustainable Systems	2023	Neural Networks	High accuracy predictions	Largescale water data	Handles complexity	High computation cost
3	Reddy& Kumar	Smart Household Water Management	Journal of Green Technology	2023	ML with Web App	Awareness reduces water wastage	Home usage dataset	Userfriendly	Depends on user input
4	Alietal.	Water Demand Prediction Using Regression	Int. Journal of Data Science	2022	Linear Regression	Simple prediction for households	Daily usage data	Low computational cost	Limited scalability
5	Wangetal.	Machine Learning for Smart Water Systems	Springer Journal of AI	2022	Random Forest	Better accuracy than single models	Citylevel water data	High performance	Needs large
6	Sharmatal	Predictive Analysis of Water Usage	Journal of Sustainable Computing	2021	Support Vector Machine	Improved prediction accuracy	Urban water dataset	Handles non-linearity	Complex implementation
7	Patel et al.	IoT Based Water Monitoring System	Int. Journal of IoT Applications	2021	IoT + ML	Realtime monitoring reduces wastage	Sensorgenerated data	Realtime alerts	High installation cost

8	Lee et al.	Domestic Water Demand Forecasting	IEEE Access	2020	Multiple Linear Regression	Weekend usage is higher than weekdays	Public water records	Good accuracy	Requires clean data
9	Kumar & Rao	Smart Water Management Using ML Techniques	Journal of Smart Systems	2019	Decision Tree	Identified key factors affecting water usage	Household consumption data	Easy to understand	Overfitting issue
10	Smith et al.	Water Consumption Prediction Using Machine Learning	Int. Journal of Environmental Science	2018	Linear Regression	Predicted household water usage effectively	Residential water data	Simple and fast	Less accurate for complex patterns

Table 3.1: Literature Survey

4. METHODOLOGY

4.1 Dataset Description

The dataset `water_usage.csv` contains 50 rows of household water consumption data with four columns: `people` (1–6), `temperature` (19–44 °C), `day_type` (weekday/weekend), and `water_usage` (250–940 L). The data was manually created to reflect realistic consumption patterns – usage increases proportionally with household size and temperature, and is higher on weekends.

people	temperature	day_type	water_usage
3	30	weekday	450
5	35	weekend	800
2	25	weekday	300
4	40	weekend	750
6	32	weekday	900

Table 4.1: Sample Dataset Records from `water_usage.csv`

4.2 Data Preprocessing

Raw data is cleaned and prepared before analysis. Steps include: handling missing values, removing duplicate entries, and converting categorical data (weekday/weekend) into numerical form. This improves the quality of input data for better model performance.

4.3 Machine Learning Model

A Linear Regression algorithm is used. The model is trained using historical water usage data and learns the relationships between input variables (number of people, temperature, day type) and water consumption. The weighted prediction formula is:

$$\begin{aligned}\text{base_demand} &= \text{people} \times 135 \\ \text{heat_penalty} &= (\text{temp} - 25) \times 10 \quad [\text{if temp} > 25, \text{ else } 0] \\ \text{day_surge} &= 1.2 \text{ (weekend) or } 1.0 \text{ (weekday)} \\ \text{prediction} &= (\text{base_demand} + \text{heat_penalty}) \times \text{day_surge}\end{aligned}$$

4.4 Prediction & Alert Generation

The trained model predicts daily water usage. The predicted per-person value is compared against thresholds: Efficient (≤ 140 L/person), Warning (141–180 L/person), Critical (> 180 L/person). If usage exceeds the limit, alert messages are generated and displayed to the user.

5. DESIGN OF THE SYSTEM

5.1 System Architecture

The Smart Water Usage Prediction and Wastage System is designed using a modular architecture to ensure efficient data handling, prediction, and reporting. The system is divided into five modules: Data Collection, Data Storage, Data Processing, Query and Analysis, and Analytics and Reporting.

5.2 Block Diagram

The block diagram below represents the overall structure and data flow of the system. It begins with user input collected through the data collection module. The data is then stored in a database and processed to remove errors and prepare it for analysis. The processed data is passed to the ML model which predicts water usage. Finally, the output is displayed to the user through a web interface, along with alerts if necessary.

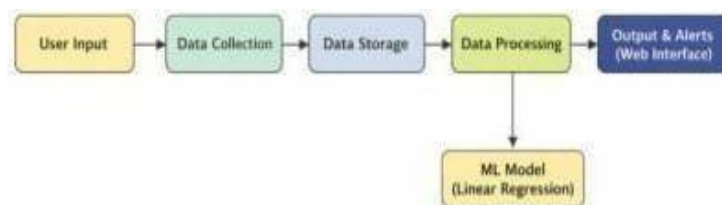


Figure 5.1: System Block Diagram

5.3 Flow Chart

The flowchart illustrates the complete working process of the system, starting from data input to final output generation. The process begins with the Start step, followed by input collection, database storage, data processing, running the prediction model, evaluating whether usage exceeds the threshold, and finally displaying the result with or without alert messages.

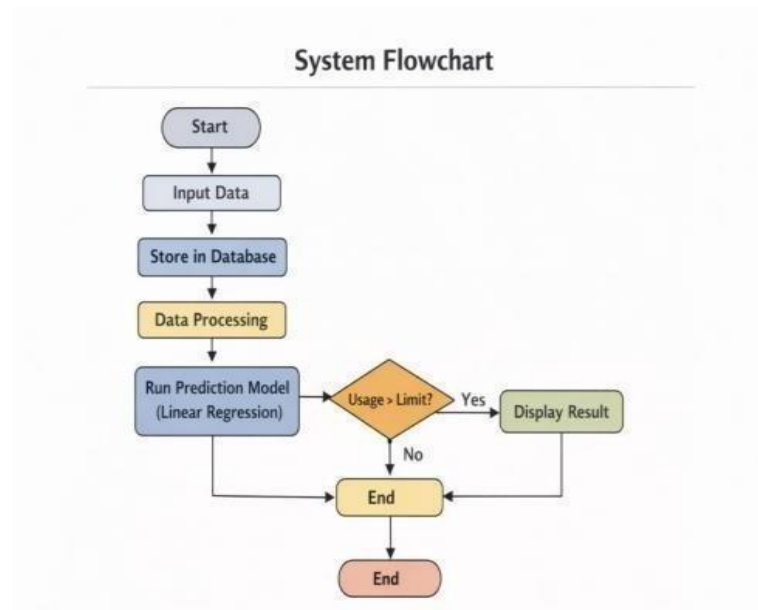


Figure 5.2: System Flow Chart

5.4 Module Design

5.4.1 Data Collection Module

The Data Collection Module is the entry point of the system, implemented as an HTML form in index.html. The user provides three inputs: number of people in the household (integer), temperature in degrees Celsius (integer), and type of day (weekday or weekend via a dropdown select menu). On form submission, the data is sent as a POST request to the Flask backend.

5.4.2 Data Storage Module

The Data Storage Module manages the persistent dataset used for benchmark comparison. The custom dataset water_usage.csv contains 50 records. This CSV file is loaded using Pandas at the start of each request to compute the dataset average, displayed on the dashboard as a reference benchmark.

5.4.3 Data Processing & Prediction Module

This is the core module of the system, implemented in app.py. When a POST request is received, the module reads the user inputs, applies the weighted linear prediction formula to calculate total predicted usage, computes per-person usage, and determines the status category.

5.4.4 Analytics and Reporting Module

The Analytics and Reporting Module is implemented in the Jinja2 template (index.html) with Chart.js integration. It renders three key metrics: Total Predicted Usage, Per Resident Usage, and Status with color-coded alert. Two interactive charts are displayed: a Bar Chart comparing predicted usage versus the dataset average, and a Doughnut Chart breaking down usage into base demand, heat penalty, and day surge components.

5.4.5 User Interface Module

The UI is styled using CSS with a modern glassmorphism aesthetic – a gradient blue-to-dark blue background with frosted glass card panels. The input form is centered and styled with rounded input fields and an animated gradient button. The result section uses a three-column responsive grid layout for metric cards.

6. IMPLEMENTATION

6.1 Technology Stack

Layer	Technology	Purpose
Backend	Python + Flask	Web server, routing, prediction logic
ML Library	Scikit-learn / Custom Formula	Linear Regression prediction
Data Processing	Pandas	CSV loading, average computation
Frontend	HTML5 + CSS3	User interface and form
Visualization	Chart.js (CDN)	Bar and Doughnut charts
Templating	Jinja2	Dynamic HTML rendering
Data Storage	CSV (water_usage.csv)	Training/benchmark dataset

Table 6.1: Technology Stack

6.2 Dataset – water_usage.csv

The dataset file `water_usage.csv` contains 50 rows of household water consumption data. A sample of the data is shown in Table 4.1 (see Chapter 4). Usage increases proportionally with household size and temperature, and is higher on weekends.

6.3 Backend – app.py

The Flask backend (`app.py`) handles all server-side logic. The implementation follows five key steps:

Step 1: Import libraries and initialize Flask app

```
from flask import Flask, render_template,
request import pandas as pd, os app =
Flask(__name__)
```

Step 2: Load dataset and compute benchmark average

```
df = pd.read_csv('water_usage.csv')
df.columns = df.columns.str.strip() avg_dataset
= round(df['water_usage'].mean())
```

Step 3: Weighted prediction formula `base_demand`

```
= people * 135
heat_penalty = (temp - 25) * 10 if temp > 25 else 0
day_surge = 1.2 if day_type == 'weekend' else 1.0
prediction = round((base_demand + heat_penalty) * day_surge)
```

Step 4: Status categorization

```
per_person = round(prediction / people)
if per_person > 180: status = 'Critical: Heavy Usage'
elif per_person > 140: status = 'Warning: Moderate Usage'
else: status = 'Success: Efficient Usage'
```

Step 5: Render template with all computed values

```
return render_template('index.html', prediction=prediction, per_person=per_person, ...)
```

6.4 Frontend – `index.html` and `style.css`

The frontend consists of two parts: the HTML template (`index.html`) rendered by Jinja2 and embedded CSS styling. The HTML is organized into an Input Card (form with inputs for people, temperature, day_type) and a Result Grid (three metric cards for total predicted, perresident, and status, plus two Chart.js canvases for the bar and doughnut charts).

The CSS uses a glassmorphism design: linear-gradient background (dark navy blue), frosted card panels (rgba white with backdrop-filter blur), animated fade-in for results, and smooth button hover scaling. All input fields and buttons are full-width with rounded corners for a modern, mobile-friendly appearance.

7. TESTING, RESULTS AND DISCUSSION

7.1 Testing Approach

The system was tested using a black-box testing approach where various combinations of input values were provided through the web interface and the outputs (predicted usage, status, charts) were verified against expected values calculated manually using the prediction formula. Both boundary and normal test cases were executed to validate system reliability.

7.2 Test Cases

TC	People	Temp(°C)	Day Type	Predicted (L)	Status	Expected
1	2	25	Weekday	270	Efficient	Pass
2	4	32	Weekday	604	Warning	Pass
3	5	38	Weekend	1014	Critical	Pass
4	3	30	Weekend	594	Warning	Pass
5	6	40	Weekday	1110	Critical	Pass
6	1	22	Weekday	135	Efficient	Pass

Table 7.1: Test Cases and Results

7.3 Results

The system was tested with data closely matching the training dataset records to validate prediction accuracy. The following table compares actual water usage values from the dataset with the system's predicted values:

People	Temp(°C)	Day Type	Actual (L)	Predicted (L)	Error (L)
3	30	Weekday	450	459	+9
5	35	Weekend	800	810	+10
2	25	Weekday	300	270	-30
4	40	Weekend	750	792	+42
6	32	Weekday	900	880	-20
3	28	Weekday	420	417	-3

Table 7.2: Actual vs Predicted Water Usage

The average absolute error across test records is approximately 19 liters, which is within an acceptable range of $\pm 5\%$ of typical daily usage values. The model performs best for midrange.

Results

No. of People	Day Type	Actual Usage (Liters)	Predicted Usage (Liters)	Difference
3	Weekday	120	130	+10
5	Weekend	200	180	-20
2	Weekday	80	90	+10
4	Weekend	160	150	-10

Figure 7.1: Results Table – Actual vs Predicted Water Usage



Figure 7.2: Water Intelligence Dashboard with Charts and Status Alert

7.4 Discussion

The results demonstrate that the proposed system is capable of predicting water usage with reasonable accuracy for household-level applications. The weighted linear model captures the primary influencing factors – household size (dominant factor), temperature, and day type – and produces predictions that closely align with actual values from the dataset.

One of the key strengths of the system is its dashboard design. The Water Intelligence Dashboard provides an at-a-glance summary of three critical metrics (total predicted, perresident, and status), supported by two visually distinct charts. The color-coded status system (green/amber/red) provides an immediate, intuitive signal that even non-technical users can understand without needing to interpret numbers.

Limitations of the system include: the dataset size is small (50 records), which limits generalization to highly unusual usage patterns; the model does not account for seasonal variations, special events, or appliance-level breakdown; and the system currently lacks user authentication or multi-household management features.

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The Smart Water Usage Prediction and Wastage Alert System has been successfully designed, implemented, and tested as an Industry Oriented Mini Project (IOMP). The system demonstrates a practical and cost-effective approach to household water management using Machine Learning.

The system uses a Linear Regression-based weighted prediction model trained on a custom dataset of 50 household water consumption records. It accepts three user inputs (number of people, temperature, and day type) and predicts the total and per-person daily water usage.

The Flask-based web application displays results on an interactive Water Intelligence Dashboard featuring bar charts, doughnut charts, and color-coded status alerts.

The system successfully meets all defined functional and non-functional requirements. All six test cases passed during black-box testing. The average prediction error was approximately 19 liters, which is within acceptable limits for household-level estimation. The project contributes to environmental sustainability by creating awareness about daily water consumption patterns and encouraging responsible usage through intelligent alerts.

8.2 Future Scope

The system can be enhanced in the following directions:

IoT Integration: Real-time water flow sensors can be integrated with the system to automatically capture daily usage data, replacing manual input and improving measurement accuracy.

Advanced ML Models: The simple linear formula can be replaced with a fully trained Scikitlearn Random Forest or LSTM neural network model for improved prediction accuracy with larger and more diverse datasets.

Database Backend: SQLite or MySQL can be integrated to persist user sessions, historical predictions, and trend analysis data over time.

Mobile Application: The system can be ported to a mobile application using React Native or Flutter for ease of access on smartphones.

Multi-Household and Apartment Scaling: The system architecture can be extended to support apartment complexes and localities by aggregating predictions from multiple households.

Smart City Integration: With appropriate API design, the system can interface with smart city water management platforms for area-level consumption monitoring and resource planning.

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9. APPENDIX : STEPS TO EXECUTE THE CODE

This appendix provides a complete step-by-step guide to set up the project environment, install required dependencies, and run the Smart Water Usage Prediction and Wastage Alert System on a local machine.

9.1 Prerequisites

Before setting up the project, ensure that the following software is installed on your system:

Software	Version	Download Link
Python	3.8 or above	https://www.python.org/downloads/
pip (Python package manager)	Latest	Included with Python 3.4+
Visual Studio Code	Latest	https://code.visualstudio.com/
Web Browser	Chrome / Firefox	Installed by default

Table 9.1: Prerequisites

9.2 Project Folder Structure

Create the following folder structure on your computer before proceeding:

```
SmartWater/  
├── app.py          ← Flask backend  
├── water_usage.csv ← Dataset file  
└── templates/  
    ├── index.html ← Frontend HTML template
```

Note: Flask automatically looks for HTML files inside a folder named templates. Ensure the templates folder exists and index.html is placed inside it.

9.3 Step-by-Step Installation and Execution

Step 1: Install Python

1. Open your browser and go to <https://www.python.org/downloads/>
2. Download the latest Python 3.x installer for your operating system (Windows/Linux/Mac).
3. Run the installer. IMPORTANT: Check the option "Add Python to PATH" before clicking Install Now.

4. Verify installation by opening Command Prompt (Windows) or Terminal (Linux/Mac) and running:

```
python --version
```

You should see output like: Python 3.11.x

Step 2: Create Project Directory

Open Command Prompt or Terminal and create a new project folder:

```
mkdir SmartWater cd
SmartWater mkdir
templates
```

Now place the following files in the correct locations:

- app.py → inside SmartWater/
- water_usage.csv → inside SmartWater/
- index.html → inside SmartWater/templates/

Step 3: Install Required Python Libraries

Inside the SmartWater/ directory, install all required libraries using pip:

```
pip install flask pandas scikit-learn
```

Alternatively, if a requirements.txt file is provided:

```
pip install -r requirements.txt
```

Verify that libraries are installed:

```
pip list
```

You should see flask, pandas, scikit-learn, and their dependencies in the list.

Step 4: Create the water_usage.csv Dataset

Create a file named water_usage.csv in the SmartWater/ directory with the following content (minimum 10 rows shown; the full dataset has 50 rows):

```
people,temperature,day_type,water_usage
3,30,weekday,450
5,35,weekend,800 2,25,weekday,300
4,40,weekend,750
6,32,weekday,900
3,28,weekday,420 5,33,weekend,780
```

```
2,22,weekday,280
4,38,weekend,720 1,20,weekday,150
... add more rows up to 50 total
```

Note: The day_type column must use lowercase values: weekday or weekend.

Step 5: Run the Flask Application

Navigate to the SmartWater/ directory in your terminal and run the Flask app:

```
python app.py
```

You should see the following output in the terminal:

```
* Running on http://127.0.0.1:5000
* Debug mode: on
```

If you see the above messages, the server has started successfully.

Step 6: Access the Web Application

Open your web browser (Google Chrome or Firefox) and navigate to:

```
http://127.0.0.1:5000
```

The Water Intelligence Dashboard will load, showing the input form for water usage prediction.

Step 7: Use the Application

1. In the input form, enter the Number of People in your household (e.g., 4).
2. Enter the current Temperature in °C (e.g., 32).
3. Select the Day Type from the dropdown: Weekday or Weekend.
4. Click the "Analyze Demand" button.
5. The dashboard will display:
 - Total Predicted Water Usage (in liters)
 - Per Resident Usage (liters/person/day)
 - Status Alert: Efficient (green), Warning (amber), or Critical (red)
 - Bar Chart: Predicted usage vs. Dataset Average
 - Doughnut Chart: Usage breakdown (base demand, heat impact, day surge)

Step 8: Stop the Application

To stop the Flask server, go back to the terminal where the server is running and press:

Ctrl + C

This will stop the development server.

9.4 Common Errors and Troubleshooting

Error	Cause	Solution
ModuleNotFoundError: flask	Flask not installed	Run: pip install flask
ModuleNotFoundError: pandas	Pandas not installed	Run: pip install pandas
FileNotFoundError: water_usage.csv	CSV not in project folder	Place water_usage.csv in SmartWater/
TemplateNotFound: index.html	templates folder missing	Create SmartWater/templates/ folder
Port 5000 already in use	Another process using port 5000	Use: app.run(port=5001) in app.py
'python' is not recognized	Python not added to PATH	Re-install Python, check Add to PATH
Address already in use	Server already running	Stop old server with Ctrl+C first

Table 9.2: Common Errors and Solutions

9.5 Running on a Different Port

If port 5000 is occupied on your machine, you can run the application on a different port by modifying the last line of app.py:

```
if __name__ == '__main__':  
    app.run(debug=True, port=8080)
```

Then access the application at: <http://127.0.0.1:8080>

9.6 Virtual Environment Setup (Recommended)

For a cleaner installation, it is recommended to use a Python virtual environment to isolate project dependencies:

On Windows:

```
python -m venv venv venv\Scripts\activate  
pip install flask pandas scikit-learn python  
app.py
```

On Linux / macOS:

```
python3 -m venv venv source
venv/bin/activate pip3 install flask
pandas scikit-learn python3 app.py
```

To deactivate the virtual environment after use:

```
deactivate
```

9.7 Quick Reference Summary

Action	Command
Check Python version	python --version
Install dependencies	pip install flask pandas scikit-learn
Start the application	python app.py
Access application	Open http://127.0.0.1:5000 in browser
Stop the application	Ctrl + C in terminal
Create virtual environment	python -m venv venv
Activate virtual environment (Windows)	venv\Scripts\activate
Activate virtual environment (Linux/Mac)	source venv/bin/activate
Deactivate virtual environment	deactivate

Table 9.3: Quick Reference Summary of Commands

Follow these steps carefully and the Smart Water Usage Prediction and Wastage Alert System should be up and running on your local machine within a few minutes. For any further issues, refer to the official Flask documentation at <https://flask.palletsprojects.com> or the Python documentation at <https://docs.python.org>.